Eulalia VoterID System — Project Completion Report

1. Project Description

Eulalia VoterID is an open-source proof of concept for digital identity and political affiliation. The original proposal addressed the risks of manual affiliation records, including misuse, falsification, and limited citizen control over personal information. Its intended ecosystem included citizens, political organizations, and electoral bodies, supported by a governance process involving relevant stakeholders.

The proposed solution was organized into four related areas:

- **Citizen and mobile layer:** a Flutter application through which citizens could manage identity information and express, or withdraw, political affiliation.
- **Integration and SSI layer:** a service boundary connecting the affiliation workflow with decentralized identity infrastructure.
- **Political-organization layer:** tools for organizations to manage affiliation information and communicate with members.
- **Electoral-body layer:** a web interface for managing political-organization information and reviewing compliance requirements.

The original technical baseline referenced Atala PRISM. The completed work is a revised, Identus-based PoC that demonstrates selected identity, authentication, SSI, and affiliation-related flows. It is not a production electoral system, an electoral certification, an institutional deployment, or a complete legal validation of the proposed model.

2. Objectives

The original objectives were to:

1. Define the stakeholders, requirements, and governance considerations for a digital political-affiliation ecosystem.
2. Implement an open-source platform with citizen, authentication, affiliation, biometric, SSI, and organization-request capabilities.
3. Provide web and mobile clients for the principal actors in the proposed workflow.
4. Integrate the affiliation workflow with the SSI technology identified in the proposal.
5. Produce and validate a proof of concept, technical documentation, updated source code, and prepare and submit an article describing the study.

The final technical evidence supports completion of a revised PoC, not full equivalence with every original Atala PRISM commitment. The completed implementation includes a .NET 8/PostgreSQL backend, a React/Vite web client, a Flutter mobile application, and an Identus-based SSI validation environment. The evidence does not establish production readiness, adoption by political organizations or electoral bodies, certification, or complete legal compliance in any jurisdiction.

3. Methodology and Activities

The project followed an iterative implementation-and-validation approach similar to the PoC model used for technical feasibility projects. Work was divided into application layers and infrastructure boundaries so that each part could be developed, exercised, and documented independently.

The main activities were:

- Refining the citizen, authentication, political-affiliation, biometric, organization-request, and SSI workflows.
- Building and connecting the .NET 8/PostgreSQL backend with the React/Vite web and Flutter mobile clients.
- Configuring the revised SSI environment with Identus Cloud Agent 1.39.0 and its Docker support stack, including APISIX, the PRISM node, PostgreSQL, Vault, and Swagger UI.
- Exercising registration, authentication, invitation, webhook, credential-status, credential-issuance, and affiliation-validation paths.
- Running component, API-integration, and SSI end-to-end tests.
- Consolidating implementation evidence, test results, coverage figures, limitations, and repository links for technical review.

The project also retained the original stakeholder and governance context: issuers, holders, verifiers, political organizations, citizens, electoral authorities, and policymakers have different responsibilities and information needs. The PoC does not replace the consultation, governance definition, legal analysis, or institutional agreements required before any real deployment.

4. Results

The final PoC and its supporting evidence provide the following results:

- Integrated backend, web, mobile, and SSI application layers for the target identity and political-affiliation scenario.
- Tested citizen registration and authentication paths.
- A documented SSI sequence from invitation through webhook and status handling to credential issuance.
- Validation controls that prevent political affiliation from proceeding without the required prerequisite credential state and related conditions.
- An Identus-based Docker validation environment. The tested `/cloud-agent/_system/health` endpoint returned Cloud Agent version `1.39.0`.
- Updated open-source repositories and integration-validation evidence suitable for technical review.

The consolidated testing evidence reports **147 tests: 144 passed and 3 failed or conditional, reported as 98.0%**. The breakdown is:

Scope	Result
Backend	132 tests; 130 passed; 2 failed
Web frontend	3/3 passed
Flutter mobile	1/1 passed
API integration	6 tests; 5 passed; 1 conditional validation
SSI end to end	5/5 passed

Backend coverage was **12.4% for lines** and **25% for branches**. These figures, the two backend failures, and the conditional API validation are part of the completion evidence. They support PoC completion within the revised technical scope, but they do not demonstrate comprehensive quality, security, performance, privacy, or production readiness.

5. Challenges and Solutions

SSI platform baseline transition. The initial proposal referenced Atala PRISM. During execution, that original technical path was no longer available or maintained for this project, so the SSI integration was adapted and validated with Identus. This was a transition in the project baseline, not a claim that the original implementation was production-ready or that the change had no consequences.

The transition affected the integration scope, the terminology used in the repositories and reports, the evidence that could be collected, and the interpretation of the original commitments. The solution was to document the revised Identus-based environment and evaluate the available evidence against that revised baseline. The final evidence therefore demonstrates the Identus-based PoC and does not claim full equivalence to every original Atala PRISM commitment.

State-dependent affiliation flow. Political affiliation depends on a required SSI credential state and related validation conditions. The solution was to preserve explicit prerequisite checks rather than accept an affiliation request solely because the application endpoint was reachable.

Integration across independent components. The backend, web client, mobile client, and SSI support services have different repositories, runtimes, and evidence types. The solution was to keep responsibilities separated and validate the layers through component, API, and end-to-end evidence.

Limited test coverage and non-passing results. The overall percentage could obscure the two backend failures, the conditional API result, and the low backend coverage. The solution was to report the detailed figures and limitations directly instead of presenting the aggregate result as a readiness claim.

Publication and milestone conditions. The planned journal-article/publication deliverable was affected by the platform transition together with other pre-established milestone conditions. This is recorded as a delivery constraint affecting the planned commitment; it is not described as a journal rejection, a direct blockage by a journal, or as simply resulting from Atala PRISM no longer being maintained. The final report consequently documents the constraint and the revised PoC evidence without asserting that the publication milestone was fulfilled.

6. Lessons Learned

- A PoC completion criterion must be distinguished from production, certification, and institutional-acceptance criteria.
- A change in technical baseline must be recorded explicitly because it changes terminology, evidence, scope, and how original commitments are interpreted.
- SSI workflows require explicit credential-state management and operational policies, not only API connectivity.
- Layered architecture makes it possible to validate application components and infrastructure separately, but integration evidence must still be reviewed as a complete flow.
- Negative and conditional test results are essential completion evidence rather than details to hide behind an aggregate percentage.
- Stakeholder governance, data protection, retention, credential lifecycle, and electoral requirements require dedicated institutional work beyond this PoC.
- A controlled PoC is useful for testing technical hypotheses, but it cannot by itself demonstrate adoption, legal validity, or readiness for a real electoral process.

7. Recommendations

1. Correct the two failing backend tests and add regression coverage for the affected validation behavior.
2. Expand backend, web, mobile, biometric, and SSI test coverage beyond the current 12.4% line and 25% branch coverage for the backend.
3. Add dedicated validation for performance, security, privacy, recovery, credential revocation, observability, and operational failure handling.
4. Define data-governance, retention, consent, credential-lifecycle, and access-control policies with the relevant stakeholders.
5. Conduct jurisdiction-specific legal and institutional analysis with political organizations and electoral bodies; do not infer legal validation from this technical PoC.
6. Clarify the revised Identus-based scope and evidence in future project plans, documentation, and publications.
7. Consider a controlled pilot only after the technical, governance, legal, privacy, and security reviews are complete and an accountable institution agrees to participate.

8. Conclusion

Eulalia VoterID completed a revised, Identus-based proof of concept within its demonstrated technical scope. The project produced an open-source integration of .NET 8/PostgreSQL, React/Vite, Flutter, and SSI infrastructure, with evidence for selected identity, authentication, credential, and political-affiliation flows. It also records the limitations required for an honest assessment: the Atala PRISM baseline transition, the publication and milestone constraints, two failing backend tests, one conditional API validation, and limited backend coverage.

The result is a technical foundation for further controlled investigation, not a production electoral system. Additional engineering, stakeholder governance, legal analysis, security review, and institutional validation are required before considering any real-world pilot.

Relevant links

- Backend (https://github.com/democracyonchain/eulalia-backend)
- Frontend (https://github.com/democracyonchain/eulalia-frontend)
- Mobile application (https://github.com/democracyonchain/eulalia-app)
- Identus-based SSI integration (https://github.com/democracyonchain/eulalia-atala-identus)
- Documentation repository (https://github.com/democracyonchain/eulalia-documents)
- Integration-validation evidence (https://github.com/democracyonchain/eulalia-documents/tree/main/integration-validation)
- Test reports (https://github.com/democracyonchain/eulalia-documents/tree/main/tests/TestIntegration)